

Mohamed Mostafa Nazih

Software Engineer

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EDUCATION

B.Sc in Computer Science, Suez Canal University

Oct 2021 – Jul 2025

- Graduation Project Grade: Excellent
- Mentored students in backend development fundamentals through Google Developer Groups.

EXPERIENCE

Oblien

Aug 2025 – Present

Founding Engineer | Part-time Remote

- Founding engineer at **Oblien**, building a programmable infrastructure layer for autonomous agents using Firecracker microVMs for secure, isolated code execution and rapid environment provisioning.
- Architected and built **Openship**, an open-source, self-hostable deployment platform with built-in CI/CD that automates application builds, deployments, routing, and TLS across desktop, web, and CLI interfaces.
- Built privacy-focused infrastructure with encrypted secrets, sandbox isolation, and strict data-ownership boundaries.
- Built a sandboxed execution platform using Firecracker microVMs to deploy and isolate agent-generated applications, ensuring secure multi-tenant execution and rapid provisioning.
- Designed and implemented backend systems powering **Reflux**, a platform that creates AI-based digital clones from users' public presence.
- Developed ingestion pipelines that processed user public data and transformed it into structured profiles used by AI agents.
- Implemented real-time interaction systems enabling users to communicate with their clones, including low-latency messaging, session handling, and stateful conversations.

Medica Scope

Jul 2025 – Present

Backend Engineer | Full-time Hybrid

- Architected and developed backend services for **CarLink**, a smart vehicle maintenance platform connecting vehicle owners with oil-change and brake service centers across Saudi Arabia, using NestJS, TypeScript, Prisma, and PostgreSQL.
- Built a modular monolith with contract- and event-driven module boundaries, supporting secure JWT/OTP authentication, geospatial service-center discovery, and end-to-end booking workflows for mobile and operator applications.
- Designed a secure OBD2 telemetry pipeline using AWS IoT Core, Kinesis, Lambda, Timestream, DynamoDB, and SNS/FCM, including X.509 device authentication, odometer tamper detection, and maintenance-threshold notifications.
- Applied Saudi PDPL requirements through regional data residency, consent tracking, personal-data deletion workflows, telemetry retention controls, and structured log redaction.
- Developed and maintained backend services for **Medica Academy**, an online medical education platform.
- Built a multi-tenant SaaS Learning Management System with microservices architecture (.NET 10.0, RabbitMQ, Redis, Firebase) handling 70+ domain services with full tenant isolation.
- Implemented event-driven communication using RabbitMQ, Redis caching/rate limiting, and Firebase Cloud Messaging for push notifications.

Onvo

May 2024 – Jul 2025

Software Engineer | Full-time, Remote

- Reworked the backend of **Onvo** from PHP to Express.js, reducing load times by 20% and improving system stability.
- Built and scaled a large Q/A platform serving 150,000+ users with real-time interactions and high concurrency support.
- Led the launch of Airwave, a platform for streaming podcasts and music, which attracted more than 10,000 users in the first week.
- Secured the backend to prevent DoS and XSS attacks, ensuring a more resilient and secure application.

PROJECTS

Aura Decor (Graduation Project) | ASP.NET, Redis, RabbitMQ, SQL Server



- AuraDecor is a modern furniture management system built with ASP.NET Core 8.0 using a clean architecture.
- It offers AI-powered text and image search, secure JWT authentication, and integrated Stripe processing.
- The system includes an advanced admin panel, RabbitMQ-based notifications, and Docker-ready deployment.

LSMSharp | C#



- Developed a high-performance LSM-Tree storage engine in C# with ACID guarantees, leveled compaction, and crash-safe WAL.
- Optimized for write-heavy workloads with Bloom filters, skip lists, and low-latency read paths.

HungerStation Microservices | .NET 8, SQL Server, Docker, Azure Service Bus, Stripe



- Designed a modular food delivery platform using Clean Architecture, DDD, CQRS, and database-per-service principles.
- Implemented event-driven communication with Azure Service Bus, JWT authentication, Stripe payments, and Dockerized deployment.

AeroUDP | Rust, Tokio, UDP, Networking



- Built a reliable asynchronous transport protocol over UDP with ordered packet delivery, retransmissions, and sliding windows.
- Implemented TCP-inspired congestion control and RTT/RTO estimation in Rust using Tokio.

SimuKernel | C#, .NET 8



- Developed an operating system simulator implementing CPU scheduling, virtual memory paging, and a mini kernel with process management.
- Supports Round Robin, Priority, and MLFQ schedulers with real-time visualization and detailed performance metrics.
- Built paging simulator with FIFO, LRU, and Optimal algorithms, providing step-by-step visualization and fault/hit statistics.
- Includes a mini in-memory kernel featuring boot sequence, CLI commands, process creation/termination, and memory allocation.

SKILLS

- **Languages & Frameworks:** C#, ASP.NET Core, JavaScript (Node.js, Express.js, NestJS, Hono, Bun), Python, C++, React Native, Next.js
- **Databases:** SQL Server, PostgreSQL, MongoDB, MySQL, MariaDB, SQLite, Redis, Memcached
- **Cloud & DevOps:** Docker, Nginx, OpenResty, Azure, AWS (EC2, S3, Lambda, ECS, API Gateway), CI/CD (GitHub Actions, Azure DevOps), Linux
- **API & Messaging:** REST, gRPC, RabbitMQ, Azure Service Bus, Stripe, Paymob
- **Testing & Tools:** xUnit, NUnit, Postman, Swagger/OpenAPI, Git
- **Core Concepts:** Clean Architecture, Microservices, Distributed Systems, CQRS, DDD, SOLID, Design Patterns, OOP, Algorithms, Data Structures, Networking